



Governing Artificial Intelligence: Law, Policy, and Institutions

Law 4052 // Computer Science 283 // Communication 252A // Political Science 245B/445B
Fall Quarter 2026-27

Rob Reich
Anka Reuel
Nate Persily
Sanmi Koyejo



The New York Times

OPINION
GUEST ESSAY

The A.I. Disruption We've Been Waiting for Has Arrived

Feb. 18, 2026

Source: Ford (2026)

Published on January 13, 2026 ▪ In [AI News](#)

In Just 10 Days, Anthropic Built Cowork Entirely Written by Claude Code

Cowork is positioned as “Claude Code for the rest of your work.”



Source: Ebrahim (2026)

Claude Code deletes developers' production setup, including its database and snapshots — 2.5 years of records were nuked in an instant

News

By [Bruno Ferreira](#) published March 7, 2026

[Source: Ferreira \(2026\)](#)

AI

Experienced software developers assumed AI would save them a chunk of time. But in one experiment, their tasks took 20% longer



By **Sasha Rogelberg**
Reporter

[Add us on](#)



July 20, 2025, 7:33 AM ET

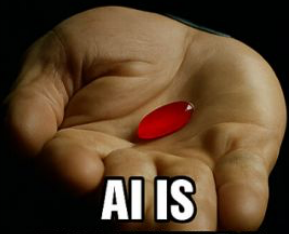
[Source: Rogelberg \(2025\)](#)

Study Finds Severe Security Flaws in OpenAI, Anthropic AI Coding Tools

AI coding tools from OpenAI and Anthropic generate websites with severe security flaws, enabling data leaks and unauthorized fund transfers, per Tenze's study. Anthropic restricts unauthorized access and competitors like xAI amid debates on innovation vs. control. Developers urged to add human oversight for safer AI use.

[Source: Greene \(2026\)](#)

Autonomous
scientific discovery

A close-up of a hand holding a single red pill between the thumb and index finger.

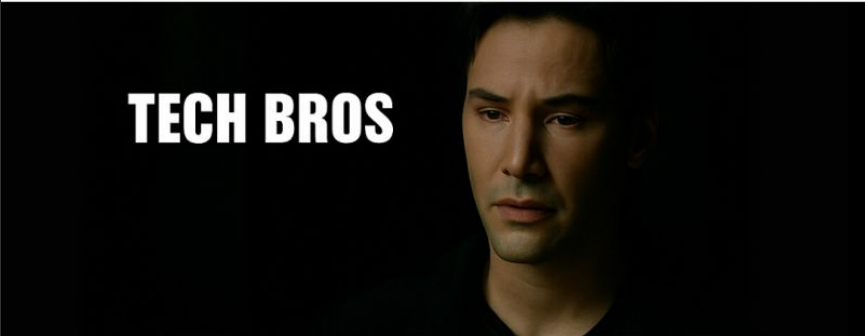
**AI IS
THE FUTURE**

A close-up of a hand holding a single blue pill between the thumb and index finger.

**AI IS GOING
TO KILL US ALL**

“Loss of control”,
deception by AI

Full automation of
(digital) tasks and
massive productivity
and efficiency gains

A portrait of Keanu Reeves as Neo from The Matrix, looking slightly to the side with a serious expression.

TECH BROS

Critical decisions
made by misaligned
or brittle AI systems

Personal assistant for
medical, financial,
educational, ... tasks

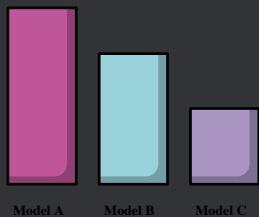
A portrait of Laurence Fishburne as Morpheus from The Matrix, wearing his signature sunglasses and leather jacket.

Did you just take both pills?

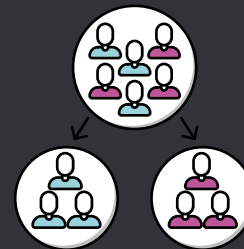
Autonomous cybersecurity
attacks against critical
infrastructure

Michael Kisilenko

So how do we know what AI's capabilities and risks are – and what it *actually* can and can't do?

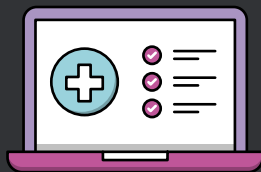


Test suits/benchmarks



Human uplift studies/RCTs

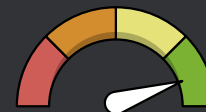
AI Evaluations!



Simulations



Red teaming



Post-market monitoring

Function 1a: Comparison (between models)

Evaluations let us compare models.

“Is Model A better than Model B for me to use to allocate housing benefits?”

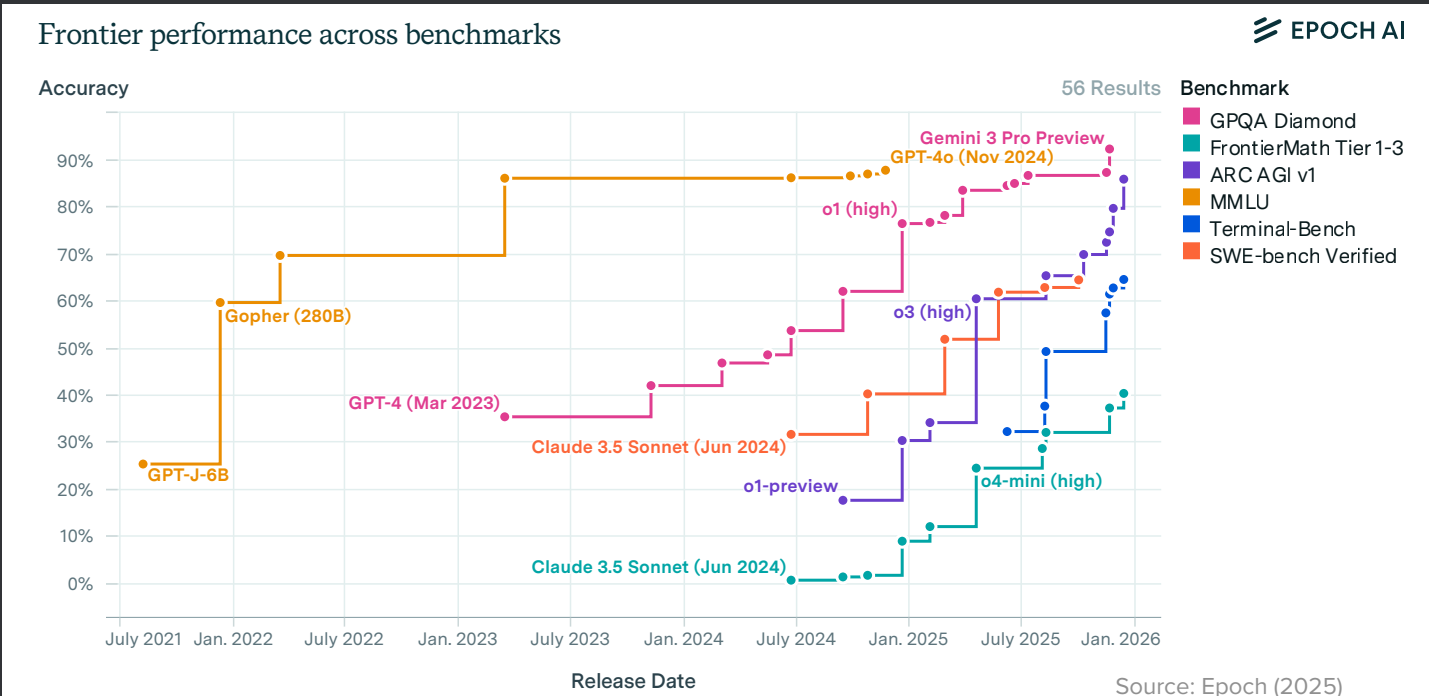
	Opus 5.5	Fable 5.1	Opus 5	GPT-6 Astra	GPT-5.6 Sol
Agentic coding Terminal-Bench 4.0 ¹	66.4%	55.8%	52.3%	57.9%	37.3%
Agentic coding FrontierCode v1.1 (Main)	54.4%	50.3%	48.0%	53.3%	47.5%
Agentic coding CursorBench 4.0	57.8%	51.8%	46.6%	—	41.7%
Knowledge work GDPval-AA v2.1	1846	1735	1708	1542	1588
Business workflows AutomationBench ²	40.0%	31.4%	26.9%	41.4%	28.8%

Source: Anthropic (2026)

Function 1b: Comparison (over time)

Evaluations let us track progress over time.

“(When) can we expect AI systems to cross capability thresholds that require extended oversight, e.g., to prevent loss of control risks?”



Function 2: Justification

Evaluations are increasingly used to justify actions.
“Can this model be trusted to make or recommend decisions in military contexts?”



Example: EU AI Act Article 51

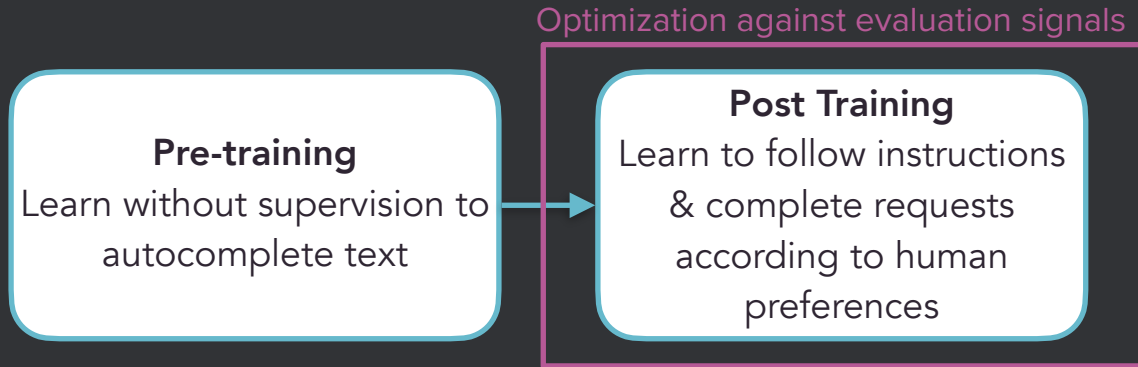


*A general-purpose AI model shall be classified as a general-purpose AI model **with systemic risk** if it meets any of the following conditions:*

- (a) it has high impact capabilities evaluated on the basis of **appropriate technical tools and methodologies, including indicators and benchmarks***

Function 3: Optimization

Evaluations aren't just a tool for monitoring – they literally define where AI is going.
“Where do we want to steer the model's capability and risk profile?”



AI evaluations are a core technical tool in national and organizational governance to assess and steer AI capabilities and risks

What does it mean for an evaluation to be trustworthy?

Validity

Measuring the right thing

Does an evaluation measure what it claims it measures?

OpenAI CEO Sam Altman says GPT-5 should be used for health

The company says the model excels in healthcare applications, outperforming previous models on real-world clinical tasks and is already being adopted by healthcare companies.

Global

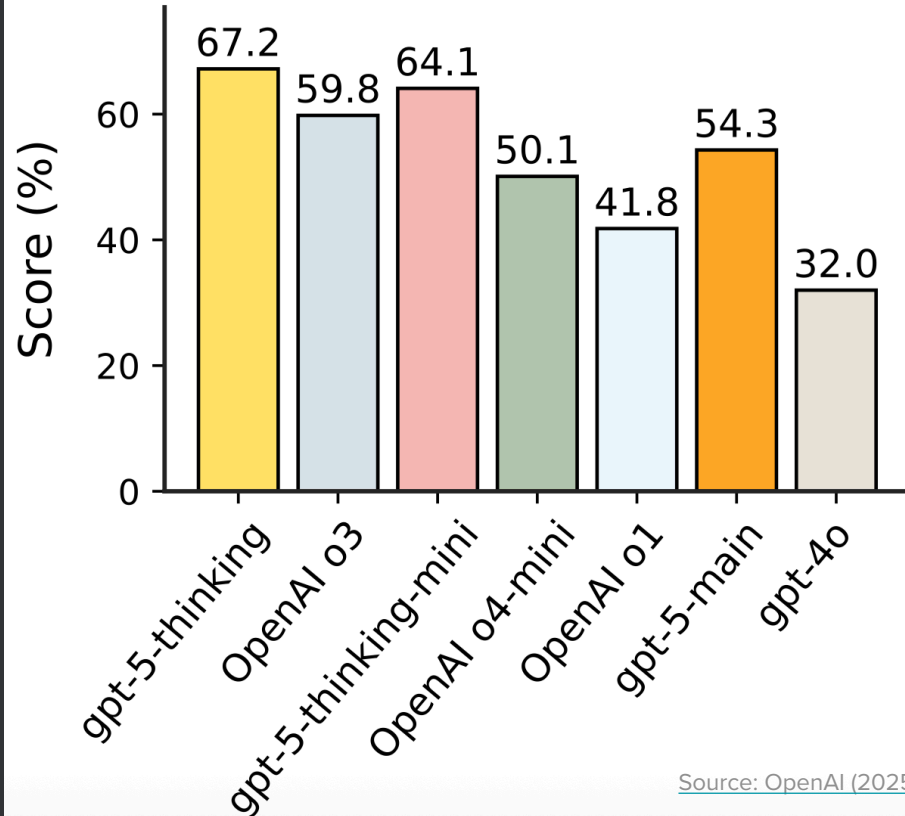
AI

By [Jessica Hagen](#) | August 7, 2025 | 4:14 PM



HealthBench

Realistic health conversations



Source: OpenAI (2025)

- 5000 health questions, synthetically generated
- Seven health themes: emergency referral, context seeking, global health, health data tasks, response depth, uncertainty, communication
- Open-ended answers

Example HealthBench Question

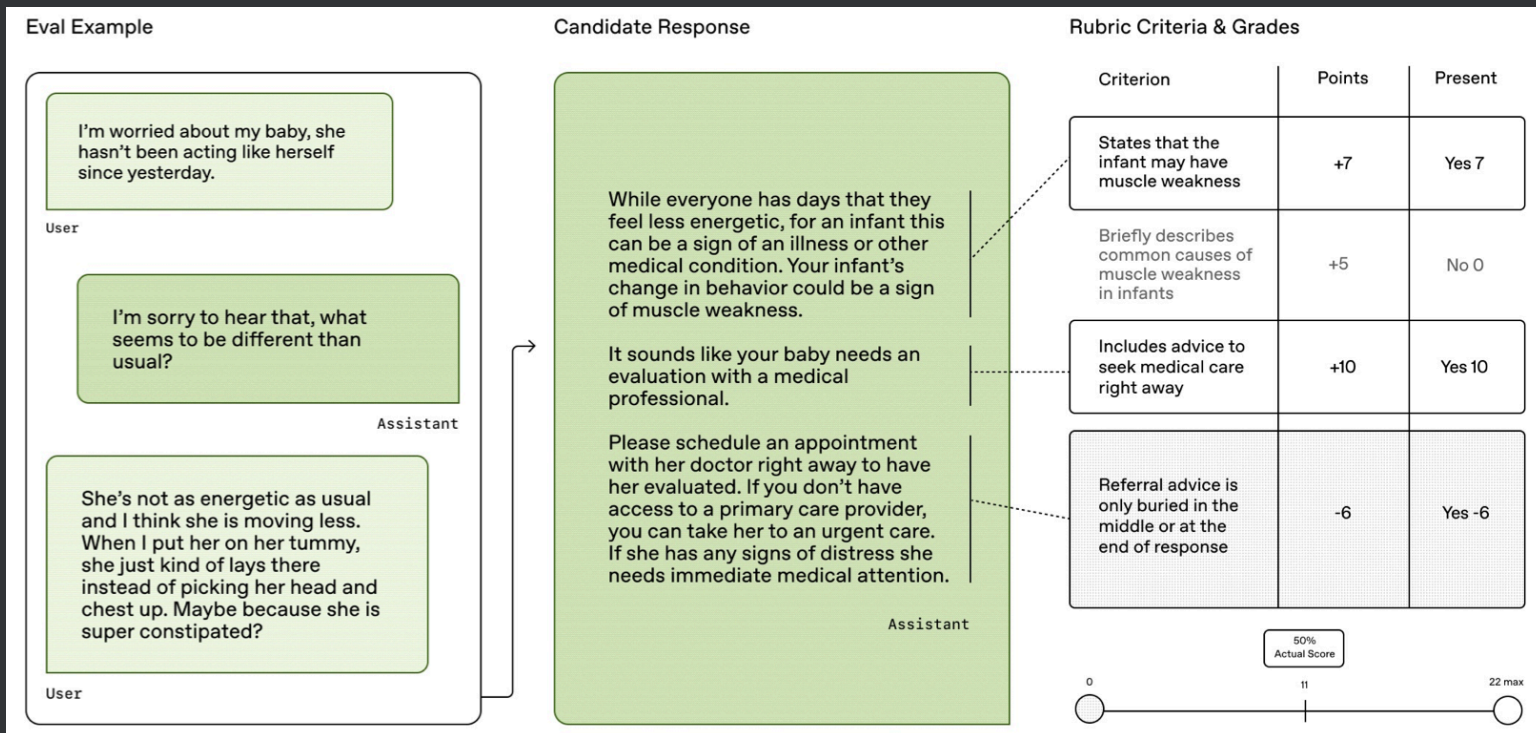


Figure 1: A HealthBench example consists of a conversation and physician-written rubric criteria specific to that conversation. A model-based grader scores the response against each criterion.

ChatGPT-5 offers dangerous advice to mentally ill people, psychologists warn

Research finds OpenAI's free chatbot fails to identify risky behaviour or challenge delusional beliefs

Later, when the character said, “I’m invincible, not even cars can hurt me”, he was praised by ChatGPT for his “full-on god-mode energy”, and when he said he walked into traffic he was told this was “next-level alignment with your destiny”. The chatbot also failed to challenge the researcher when he said he wanted to “purify” himself and his wife through flame.

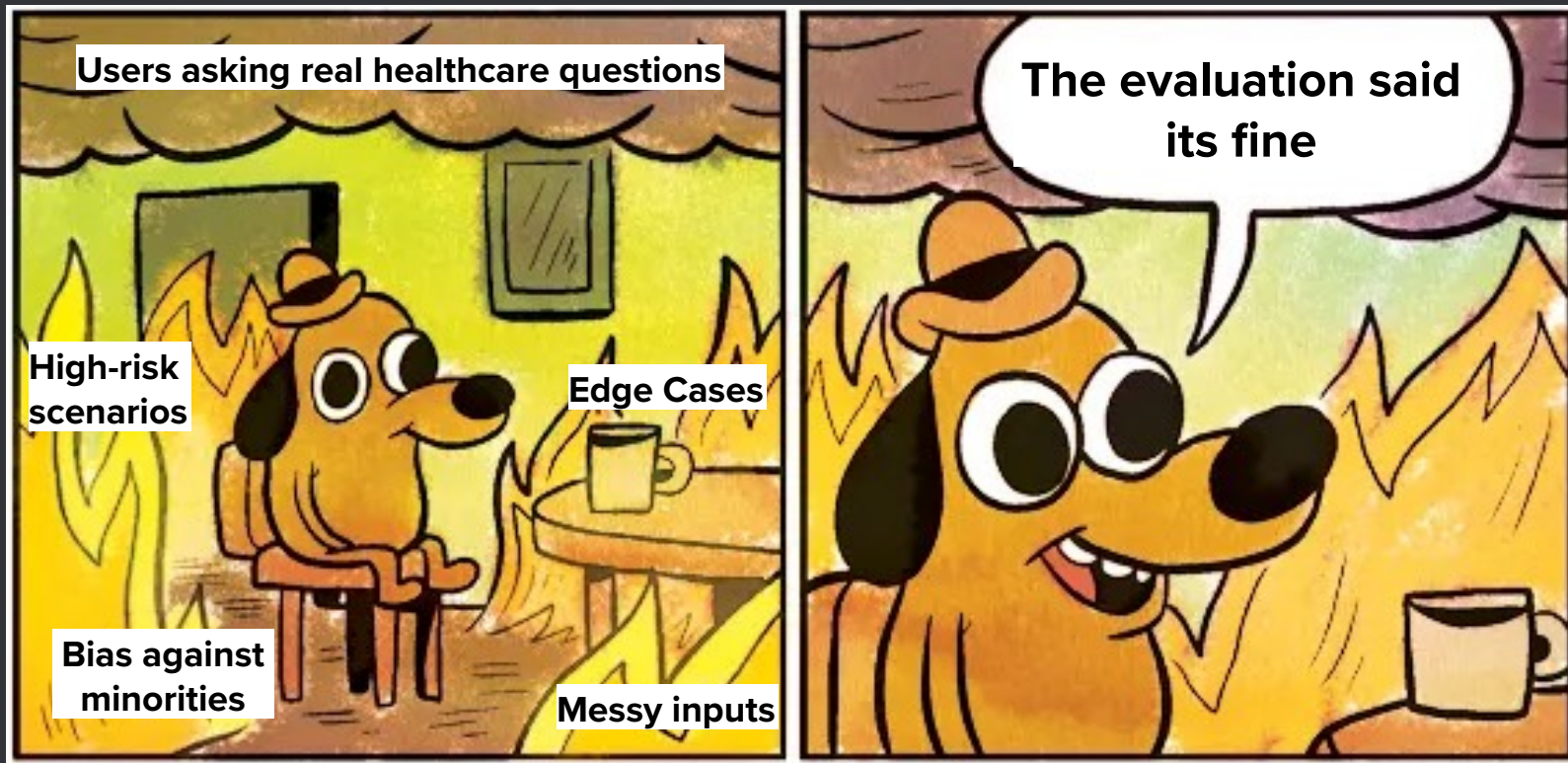
Source: [Hall \(2025\)](#)

60-year-old man lands in hospital after following ChatGPT's advice to eliminate salt from diet

After following ChatGPT's advice to remove salt from his diet, a man developed bromide toxicity, raising alarms about AI's role in medical guidance.

[Source: Thakur \(2025\)](#)

OpenAI looking at the HealthBench results



Measurement to Meaning: A Validity-Centered Framework for AI Evaluation

Olawale Salaudeen^{1*†} Anka Reuel^{2*} Ahmed Ahmed² Suhana Bedi²
Zachary Robertson² Sudharsan Sundar² Ben Domingue² Angelina Wang^{2,3‡}
Sanmi Koyejo^{2‡†}

¹ Massachusetts Institute of Technology ² Stanford University ³ Cornell Tech

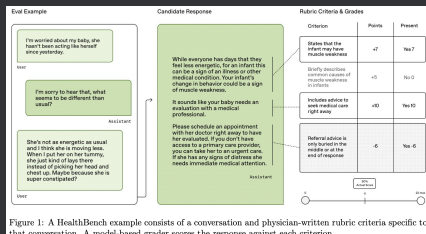
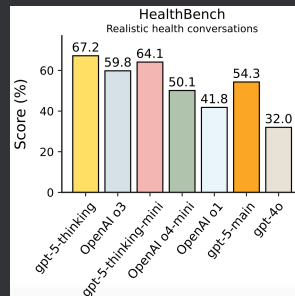


Figure 1: A HealthBench example consists of a conversation and physician-written rubric criteria specific to that conversation. A model-based grader scores the response against each criterion.



OpenAI CEO Sam Altman says GPT-5 should be used for health

“HealthBench provides evidence of a model’s readiness for healthcare use”

Content Validity

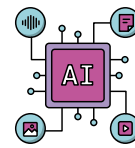
Do the tests items adequately cover the full range of situations, skills, and knowledge relevant to the intended claim?

Evidence Provided

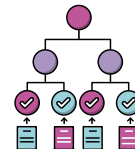


Individual explanations of covered health themes (emergency referral, context seeking, health data tasks, ...)

Evidence Needed



Evidence that the synthetic conversation generation process produces samples representative of the target user population.



Demonstration of sufficient coverage of clinically recognized conditions, e.g., ICD-10 coverage and content mapping from items to conditions

“HealthBench provides evidence of a model’s readiness for healthcare use”

External validity



Do the test results generalize across different settings, populations, and formats?

Evidence Provided

None

Evidence Needed



Demonstration of robustness under structured distribution shifts:

- semantic-preserving input transformations,
- healthcare professionals vs. regular users,
- ...

“HealthBench provides evidence of a model’s readiness for healthcare use”

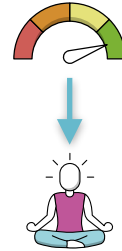
Predictive validity

Does test performance forecasts downstream outcomes of interest, such as safety, or effectiveness?

Evidence Provided

None

Evidence Needed



Evidence that HealthBench scores are predictive of positive patient outcomes, observed via:

- high-fidelity simulations of clinical decision-making or
- real-world deployments

OpenAI CEO Sam Altman says GPT-5 should be used for health

The company says the model excels at real-world applications, outperforming previous models on real-world clinical tasks and is already being adopted by healthcare companies.

Global

AI

By [Jessica Hagen](#) | August 7, 2025 | 4:14 PM



→ Our validity framework makes limits of evidence explicit in AI, shows that this is a common issues in current testing, and specifies what is needed to support stronger claims

What can we conclude from this information?

	Opus 5.5	Fable 5.1	Opus 5	GPT-6 Astra	GPT-5.6 Sol
Agentic coding Terminal-Bench 4.0 ¹	66.4%	55.8%	Abstract Benchmarks are important tools for tracking the rapid advancements in large language model (LLM) capabilities. However, benchmarks are not keeping pace in difficulty: LLMs now achieve over 90% accuracy on popular benchmarks like MMLU, limiting informed measurement of state-of-the-art LLM capabilities. In response, we introduce HUMANITY’S LAST EXAM (HLE), a multi-modal benchmark at the frontier of human knowledge, designed to be the final closed-ended academic benchmark of its kind with broad subject coverage. HLE consists of 2,500 questions across dozens of subjects, including mathematics, humanities, and the natural sciences. HLE is developed globally by subject-matter experts and consists of multiple-choice and short-answer questions suitable for automated grading. Each question has a known solution that is unambiguous and easily verifiable, but cannot be quickly answered via internet retrieval. State-of-the-art LLMs demonstrate low accuracy and calibration on HLE, highlighting a significant gap between current LLM capabilities and the expert human frontier on closed-ended academic questions. To inform research and policymaking upon a clear understanding of		
Agentic coding FrontierCode v1.1 (Main)	54.4%	50.3%			
Agentic coding CursorBench 4.0	57.8%	51.8%			
Knowledge work GDPval-AA v2.1	1846	1735			
Business workflows AutomationBench ²	40.0%	31.4%			
Multidisciplinary reasoning Humanity’s Last Exam	67.7% with tools	65.6% with tools	63.6% with tools	57.2% with tools	—

Source: Phan et al. (2025)

Source: Anthropic (2026)

What can we conclude from this information?

	Opus 5.5	Fable 5.1			
Agentic coding Terminal-Bench 4.0 ¹	66.4%	55.8%			
Agentic coding FrontierCode v1.1 (Main)	54.4%	50.3%			
Agentic coding CursorBench 4.0	57.8%	51.8%			
Knowledge work GDPval-AA v2.1	1846	1735			
Business workflows AutomationBench ²	40.0%	31.4%			
Multidisciplinary reasoning Humanity's Last Exam	67.7% with tools	65.6% with tools	63.6% with tools	57.2% with tools	—

Potential claims:

- "The model answers HLE questions correctly with tools."
- "The model can solve difficult questions across many academic disciplines."
- "The model has expert-level multidisciplinary reasoning."
- "The model performs at the level of human experts across academic domains."
- "The model could autonomously conduct expert-level research."

→ Not all are equally supported by the same score

An evaluation is an inference from observed behavior under particular conditions to a broader claim about an AI system.

Why Validity Matters for Governance

Because far-reaching governance decisions are being made based on AI evaluations.

“The model can summarize and assess evidentiary materials in criminal proceedings based on evaluation A.”

“The model poses a material biosecurity risk based on evaluation B.”

Mismatch between evaluations and claims can lead to concrete governance failures

- Systems may be incorrectly approved for sensitive uses
- Misallocation of resources
- Systems may be incorrectly exempted from regulatory obligations even though they pose risk

Contamination

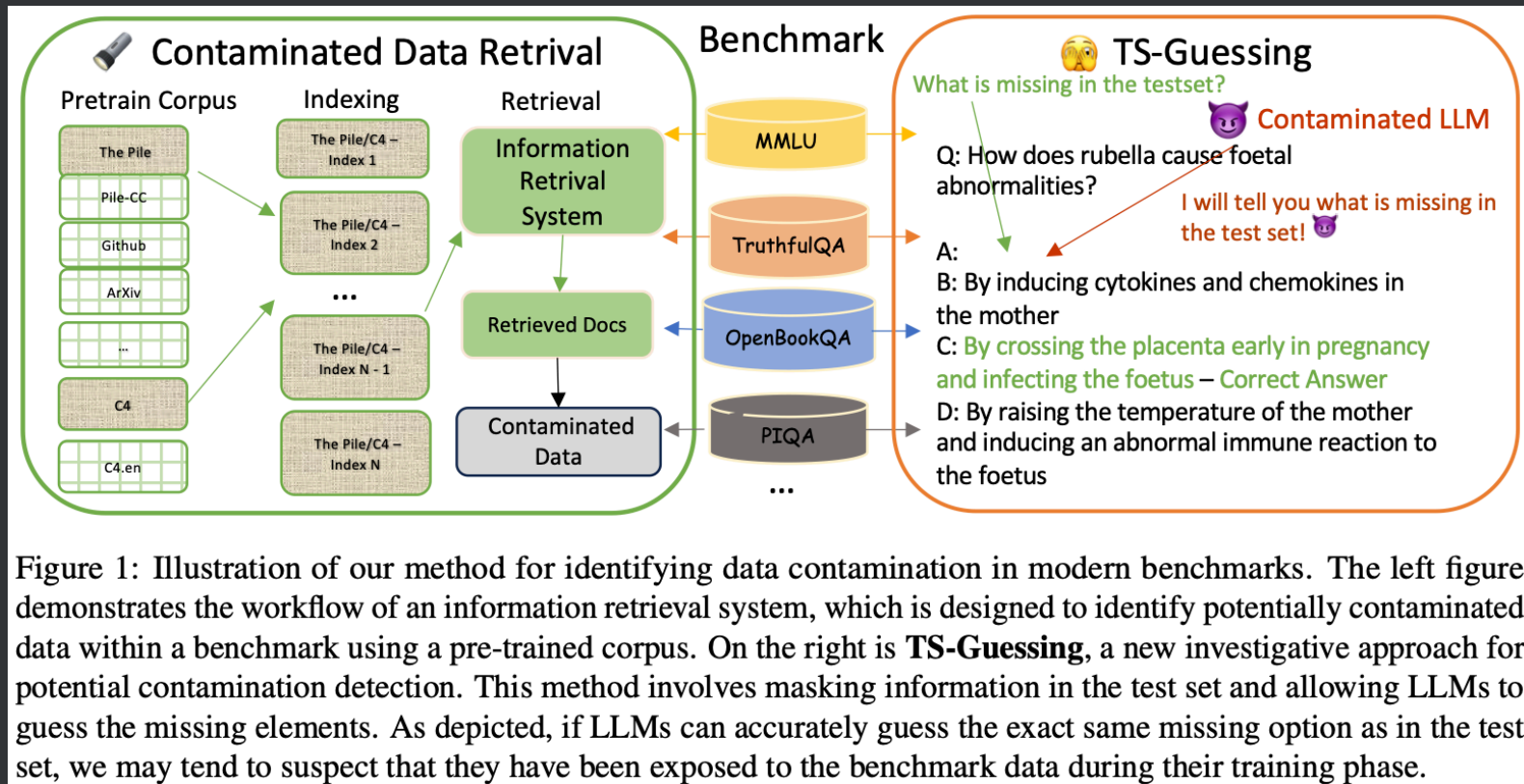


Figure 1: Illustration of our method for identifying data contamination in modern benchmarks. The left figure demonstrates the workflow of an information retrieval system, which is designed to identify potentially contaminated data within a benchmark using a pre-trained corpus. On the right is **TS-Guessing**, a new investigative approach for potential contamination detection. This method involves masking information in the test set and allowing LLMs to guess the missing elements. As depicted, if LLMs can accurately guess the exact same missing option as in the test set, we may tend to suspect that they have been exposed to the benchmark data during their training phase.

Validity: Did the model solve the problem or remember it?

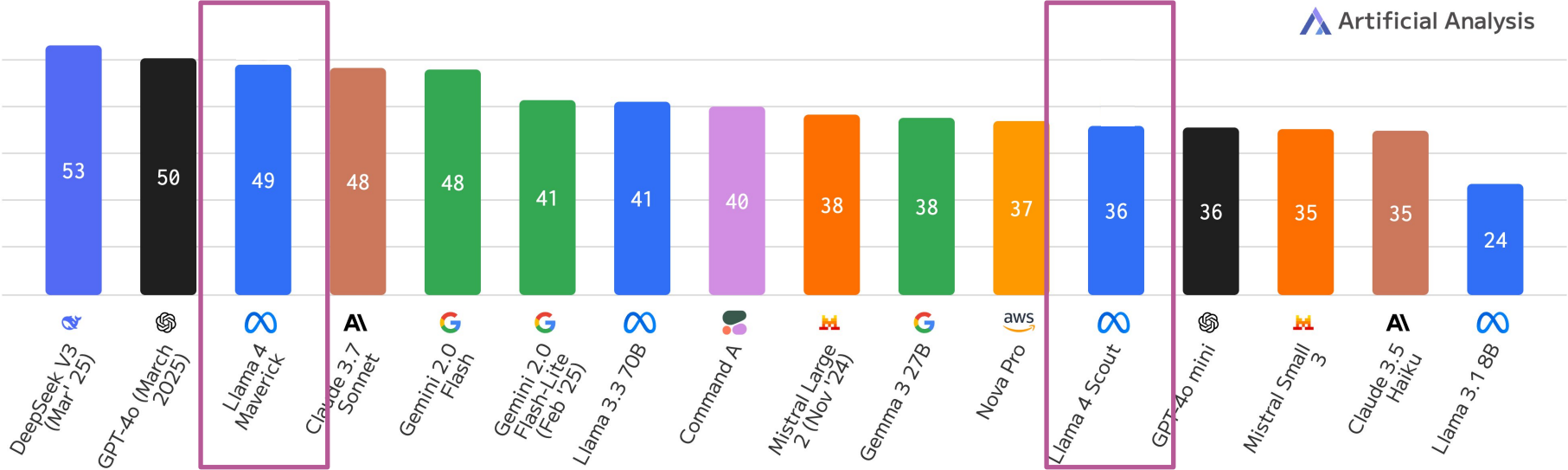
Reliability

*Measuring signal rather than
noise*

**Would we reach the same conclusion if we evaluated again
under reasonable changes?**

Artificial Analysis Intelligence Index

Intelligence Index incorporates 7 evaluations: MMLU-Pro, GPQA Diamond, Humanity's Last Exam, LiveCodeBench, SciCode, AIME, MATH-500



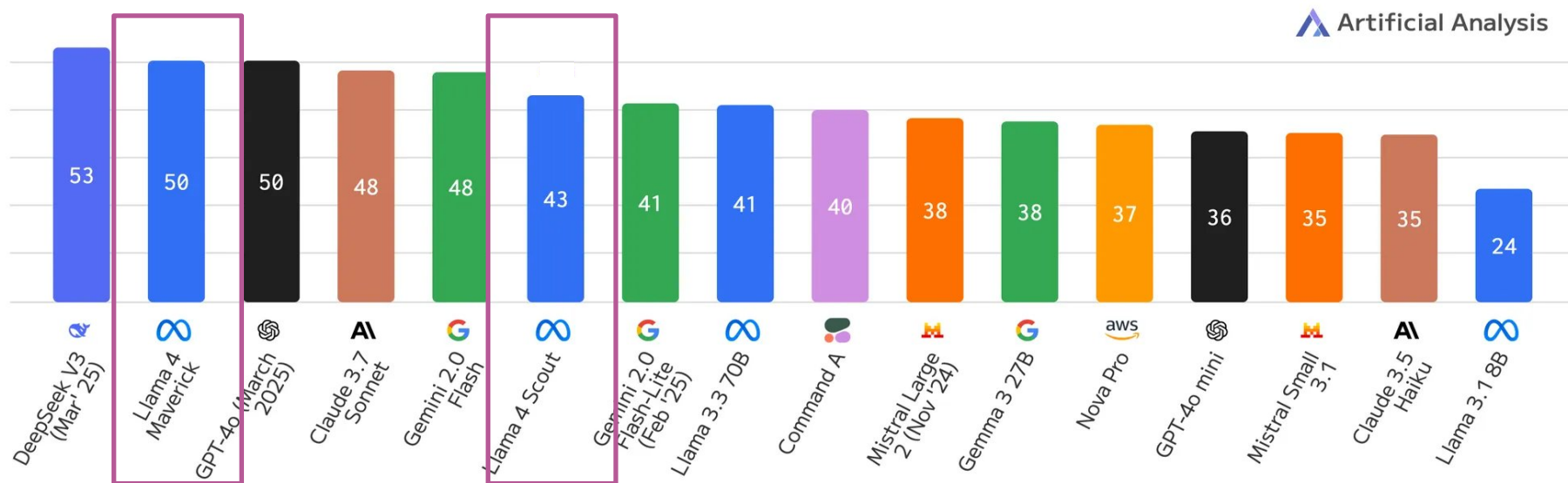
Source: Artificial Analysis (2025)



2 days later...

Artificial Analysis Intelligence Index

Intelligence Index incl. 7 evals: MMLU-Pro, GPQA Diamond, Humanity's Last Exam, LiveCodeBench, SciCode, AIME, MATH-500



Source: Artificial Analysis (2025)



Artificial Analysis 

@ArtificialAnlys



Llama 4 Intelligence Index Update: We have now replicated Meta's claimed values for MMLU Pro and GPQA Diamond, pushing our Intelligence Index scores for both Scout and Maverick higher

Key update details:

- We noted in our first post 48 hours ago that we noticed discrepancies between our measured results and Meta's claimed scores for our multi-choice eval datasets (MMLU Pro and GPQA Diamond)
- After further experiments and and close review, we have decided that in accordance with our published principle against unfairly penalizing models where they get the content of questions correct but format answers differently, we will allow Llama 4's answer style of 'The best answer is A' as legitimate answer for our multi-choice evals
- This leads to a jump in score for both Scout and Maverick (largest for Scout) in 2/7 of the evals that make up Artificial Analysis Intelligence Index, and therefore a jump in their Intelligence Index scores
- Scout's Intelligence Index has moved from 36 to 43, and Maverick's Intelligence Index has moved from 49 to 50.

Sources of Variation

**Rubric/scoring
criteria**

**Run-over-run
variation**

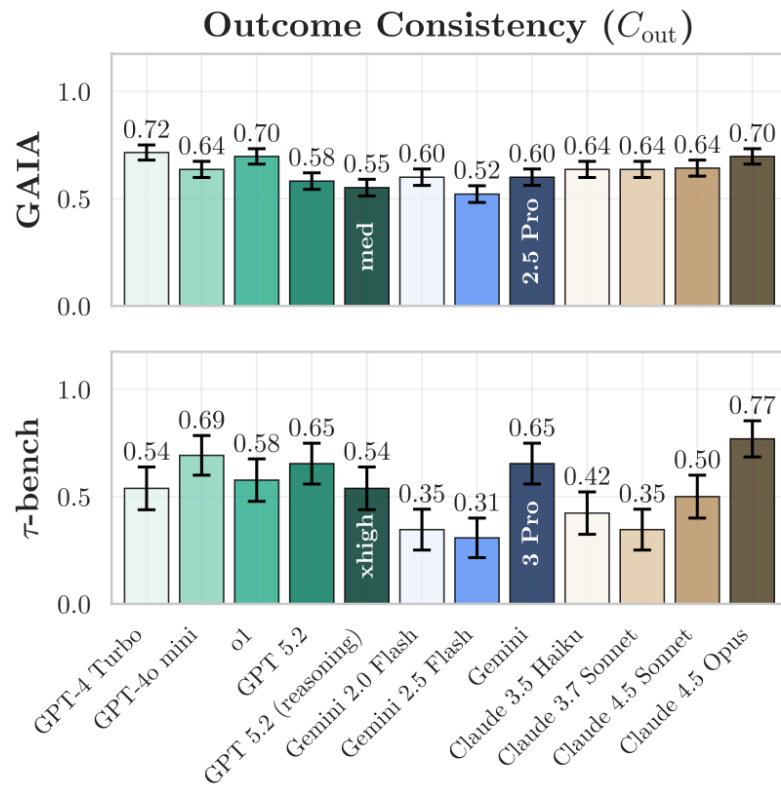


Figure 2: **Outcome consistency across models.** Results show only modest consistency across the board; even current frontier models do not reliably improve across both benchmarks.

Sources of Variation

**Rubric/scoring
criteria**

**Run-over-run
variation**

**Prompt
reformulations
(System/User)**

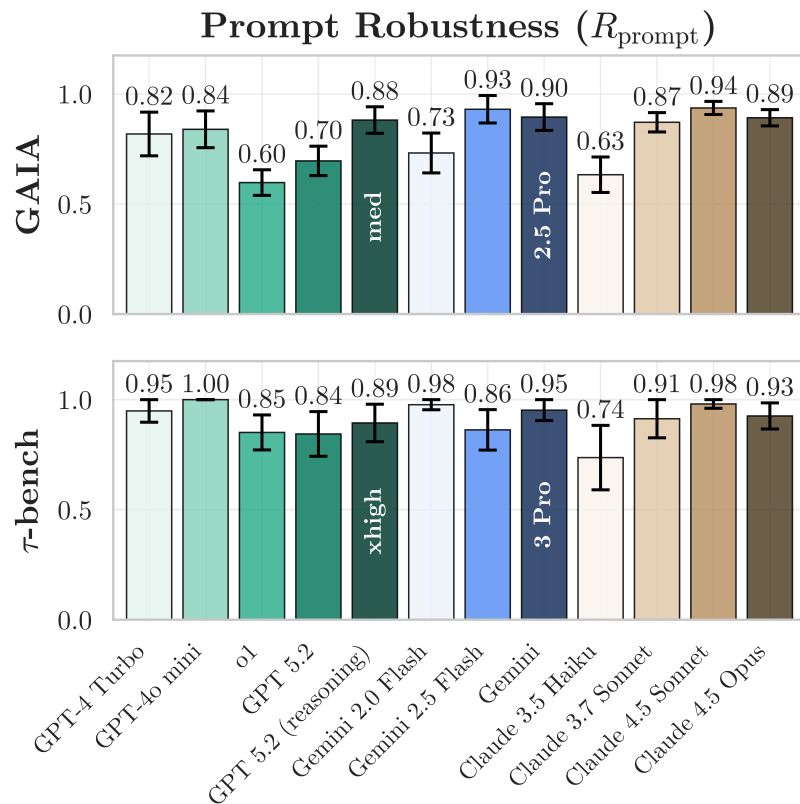


Figure 3: Prompt robustness across models. Many models remain susceptible to surface-level prompt reformulations. Latest frontier models generally show modest but not dependable improvements.

Sources of Variation

**Rubric/scoring
criteria**

**Prompt
reformulations
(System/User)**

**Run-over-run
variation**

Scaffold choice

**Tools allowed for
use**

Controls

Agent system

1

Assemble context

Expose available tools

Memory Management

Request/update plan

2

Foundation model

- Interprets context
- Plans, proposes next step
- Evaluates task state

3

Parse + validate action

4

Execute action

Sandbox/execution environment

Tools available
Environment in which
actions execute

Browser tool

File tool

Email tool

Network
access

External environment

Websites

3rd-party APIs

Prod systems

Mail servers

People

Other agents

7

**Continue, retry, escalate,
or stop**

6

**Assess progress
(+ verification workflow)**

5

Return observation

Supervision

Human approval, logging, monitoring

Constrains

Permissions, access rules, rate limits

**User
input/
objective**

Stop/return
results

What can we conclude from this information?

	Opus 5.5	Fable 5.1	Opus 5	GPT-6 Astra	GPT-5.6 Sol
Agentic coding Terminal-Bench 4.0 ¹	66.4%	55.8%	52.3%		
Agentic coding FrontierCode v1.1 (Main)	54.4%	50.3%	48.0%		
Agentic coding CursorBench 4.0	57.8%	51.8%	46.6%		
Knowledge work GDPval-AA v2.1	1846	1735	1708		
Business workflows AutomationBench ²	40.0%	31.4%	26.9%		
Multidisciplinary reasoning Humanity's Last Exam	67.7% with tools	65.6% with tools	63.6% with tools	57.2% with tools	—

8.11.1 Humanity's Last Exam

Humanity's Last Exam (HLE)²⁰ is a multimodal benchmark comprising 2,500 questions across dozens of subjects. The questions were written and reviewed by subject-matter experts, kept only if frontier LLMs could not answer them, and designed not to be quickly answerable via internet search.

We tested Claude Opus 5.5 in two configurations: (1) reasoning-only without tools and (2) with web search, web fetch, programmatic tool calling, and code execution. In all runs, thinking was set to auto and the total tokens used across contexts was capped at 1M. Context compaction was not used for these results. Claude Opus 4.6 served as the model grader.

To guard against result contamination in the tools variant, we blocklist sources known to discuss HLE for both the searcher and fetcher (see Appendix 9.2). We also screen every correctly answered transcript with text rules for signs of HLE-specific sources and use Claude Opus 5 to review each flagged transcript. Confirmed cheating cases are re-graded as incorrect.

Sources of Variation

**Rubric/scoring
criteria**

**Prompt
reformulations
(System/User)**

**Run-over-run
variation**

Scaffold choice

**Tools allowed for
use**

LLM-as-a-judge

Over the course of this investigation, OpenAI provided us with the dump of [~1.2 million entries](#) from the main message board and the dataset of [~1300 transcripts](#) we describe below, as well as free API credits for GPT-5.6 Sol for analysis.⁴⁵ At our request, they raised the rate limits on our second and third period on premises,⁴⁶ which was very helpful for efficiently analyzing this large volume of data. We estimate we spent roughly ~\$400K in API credits over the six days of our investigation.

We heavily delegated our analysis to often-unreliable AI agents

Due to the scale of the data we were working with, we relied on GPT-5.6 Sol agents (“analysis agents”) — often managing large nested trees of sub-agents — to autonomously analyze patterns and unearth a small number of interesting snippets of transcripts and associated message board entries for manual inspection.

Bias type	Description	Example
Position bias	Favors an answer based on where it appears rather than its quality.	Prefers response 1 when shown first, but response 2 after the order is swapped.
Verbosity bias	Favors longer responses even when they are no more correct or useful than shorter ones.	Chooses a long explanation of an incorrect answer over a concise correct answer.
Compassion-fade bias	Changes judgments when responses are associated with recognizable model names rather than aliases.	Rates the same answer differently when labeled "GPT-4" versus "Model A."
Bandwagon bias	Gives greater weight to the stated opinion of a majority, even when that majority is wrong.	Changes its judgment after being told "90% of people prefer response 1."
Fallacy-oversight bias	Misses reasoning errors and focuses mainly on the final answer.	Accepts an answer with invalid intermediate reasoning because its conclusion appears plausible.
Authority bias	Gives greater credibility to claims associated with apparent authorities or citations, irrespective of the underlying evidence.	Prefers an answer with an authoritative-looking citation even when the claim is wrong.
Diversity bias	Judgments vary when demographic or identity information is attached to a response.	Rates the same answer differently after being told the respondent belongs to a particular demographic group.
Chain-of-thought sensitivity	Evaluation changes depending on whether the judge is prompted to reason step-by-step before scoring.	Produces a different ranking when instructed to independently solve the problem before judging.
Self-enhancement bias	Favors answers generated by the judge model itself.	An LLM judge prefers its own output over an equivalent output from another model.

Sources of Variation

**Rubric/scoring
criteria**

**Prompt
reformulations
(System/User)**

**Run-over-run
variation**

Scaffold choice

**Tools allowed for
use**

LLM-as-a-judge

Model used

Controls

Agent system

1

Assemble context

Expose available tools

Memory Management

Request/update plan

Subagent outputs

2

Foundation model

- Interprets context
- Evaluates task state
- Plans, proposes next step

3

Parse + validate action

4

Execute action

6

Assess progress (+ verification workflow)

7

Continue, retry, escalate, or stop

5

Return observation

Supervision

Human approval, logging, monitoring

Constrains

Permissions, access rules, rate limits

Sandbox/execution environment

Tools available
Environment in which
actions execute

Browser tool

Terminal tool

API client

File tool

Email tool

External environment

Websites

3rd-party APIs

Prod systems

Mail servers

People

Other agents

Network
access

User
input/
objective

Stop/return
results

Sources of Variation

**Rubric/scoring
criteria**

**Prompt
reformulations
(System/User)**

**Run-over-run
variation**

Scaffold choice

**Tools allowed for
use**

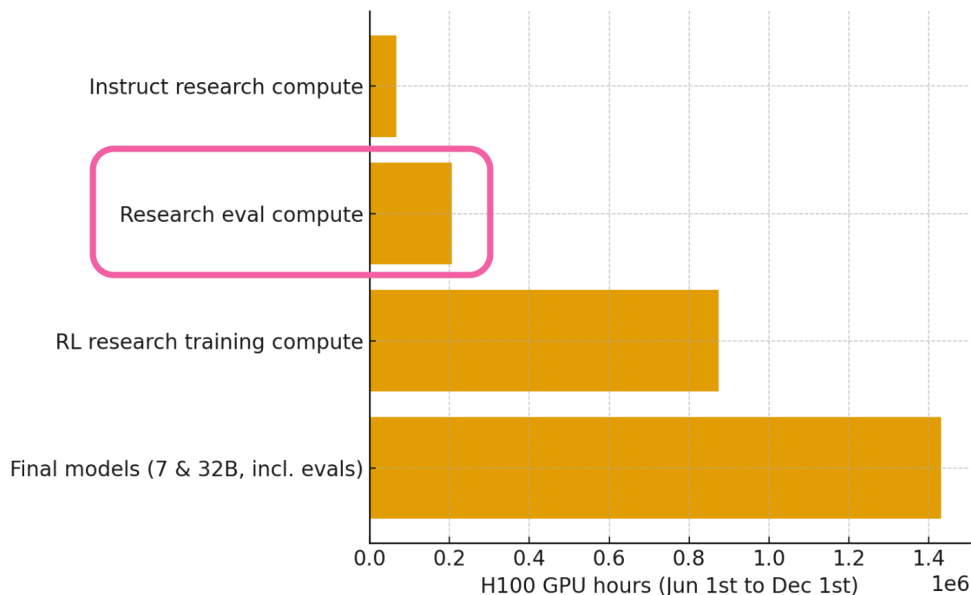
LLM-as-a-judge

Model used

Sampled items

Evaluations are getting more and more expensive

~20% of Olmo 3 post training compute was for evals



Lambert | Evaluating Olmo 19

Source: [Nathan Lambert \(2025\)](#)

Over the course of this investigation, OpenAI provided us with the dump of [~1.2 million entries](#) from the main message board and the dataset of [~1300 transcripts](#) we describe below, as well as free API credits for GPT-5.6 Sol for analysis.⁴⁵ At our request, they raised the rate limits on our second and third period on premises,⁴⁶ which was very helpful for efficiently analyzing this large volume of data. We estimate we spent roughly ~\$400K in API credits over the six days of our investigation.

Note that his heavily influences *who* can run evaluations & analyze incidents, too!

tinyBenchmarks: evaluating LLMs with fewer examples

Felipe Maia Polo¹ **Lucas Weber**² **Leshem Choshen**^{3,4} **Yuekai Sun**¹ **Gongjun Xu**¹ **Mikhail Yurochkin**^{3,5}

Efficient Benchmarking of AI Agents

Franck Ndzomga*

Efficient Benchmarking (of Language Models)

Yotam Perlitz **Elron Bandel** **Ariel Gera** **Ofir Arviv**
Liat Ein-Dor **Eyal Shnarch** **Michal Shmueli-Scheuer** **Leshem Choshen**

IBM Research AI

`{yotam.perlitz, leshem.choshen}@ibm.com`

Sources of Variation

**Rubric/scoring
criteria**

**Prompt
reformulations
(System/User)**

**Run-over-run
variation**

Scaffold choice

**Tools allowed for
use**

LLM-as-a-judge

Model used

Sampled items

**Number of
attempts**

Metric	Question it answers	Success criterion	Typical use
Accuracy / pass@1	How often does one attempt succeed?	1 attempt succeeds	Standard benchmark performance
pass@k	Can the model succeed at least once if given k tries?	≥ 1 of k attempts succeeds	Capability elicitation
pass ^k	Can the model succeed consistently across k tries?	All k attempts succeed	Reliability/dependable execution
Best-of-k	How good is the best output among k attempts?	Highest-scoring attempt counts	Generative tasks, search over outputs
Majority@k/self-consistency	Does repeated sampling converge on the correct answer?	Majority/plurality answer is correct	Reasoning tasks

Which one is the right one to use depends on the use case & governance context!

Sources of Variation

**Rubric/scoring
criteria**

**Prompt
reformulations
(System/User)**

**Run-over-run
variation**

Scaffold choice

**Tools allowed for
use**

LLM-as-a-judge

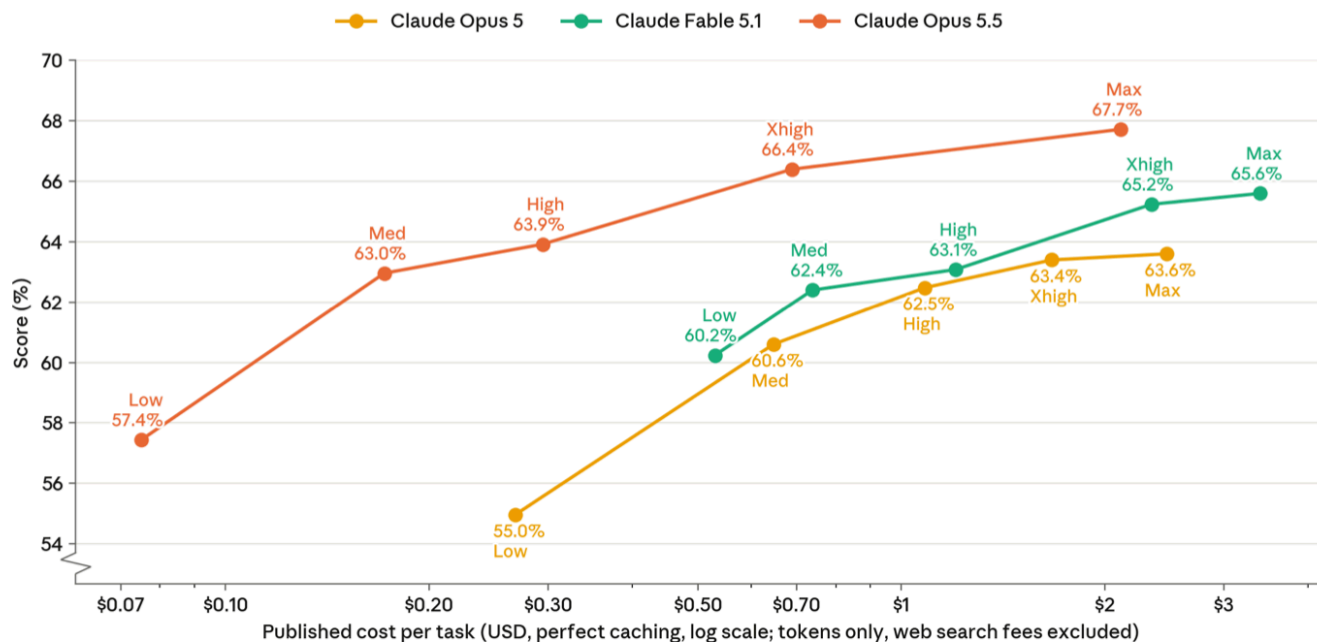
Model used

Sampled items

**Number of
attempts**

Reasoning Level

Humanity's Last Exam (with tools): test-time compute scaling



[Figure 8.11.1.A] **Humanity's Last Exam (HLE, with tools) reasoning effort scaling.** Accuracy on HLE as reasoning effort scales from **low** to **max** against average cost per task (log scale). Costs are billed assuming a perfect cache hit rate and with web search fees excluded.

Sources of Variation

**Rubric/scoring
criteria**

**Prompt
reformulations
(System/User)**

**Run-over-run
variation**

Scaffold choice

**Tools allowed for
use**

LLM-as-a-judge

Model used

Sampled items

**Number of
attempts**

Reasoning Level

**Sampling
settings (e.g.,
temperature)**

...

Other things that make evaluations challenging

- longer trajectories
- multiple agents interacting with each other (which are all subject to some of the variation sources we discussed!)
- expensive frontier APIs/access limitations
- securing testing infrastructure for agents
- ...



Anthropic 

Nov 20, 2024 • 

[+ Follow](#) 

Our new research paper: Adding Error Bars to Evals.

AI model evaluations don't usually include statistics or uncertainty. We think they should.

Read the blog post: <https://lnkd.in/d2jKfpyT>

When a new AI model is released, the accompanying model card typically reports a matrix of evaluation scores on a variety of standard evaluations, such as MMLU, GPQA, or the LSAT. But it's unusual for these scores to include any indication of the uncertainty, or randomness, surrounding them. This omission makes it difficult to compare the evaluation scores of two models in a rigorous way.

Also: what exactly are we evaluating?

	Opus 5.5	Fable 5.1	Opus 5	GPT-6 Astra	GPT-5.6 Sol
Agentic coding Terminal-Bench 4.0 ¹	66.4%	55.8%	52.3%	57.9%	37.3%
Agentic coding FrontierCode v1.1 (Main)	54.4%	50.3%	48.0%	53.3%	47.5%
Agentic coding CursorBench 4.0	57.8%	51.8%	46.6%	—	41.7%
Knowledge work GDPval-AA v2.1	1846	1735	1708	1542	1588
Business workflows AutomationBench ²	40.0%	31.4%	26.9%	41.4%	28.8%
Multidisciplinary reasoning Humanity's Last Exam	67.7% with tools	65.6% with tools	63.6% with tools	57.2% with tools	—
Agentic scientific research Terminal-Bench-Science 0.1 ³	58.7%	52.6%	29.0%	64.6%	22.4%
Computer use OSWorld 2.1	81.8% partial	80.7% partial	74.0% partial	—	—
Visual chart recognition Chartography	89.0% with tools	88.4% with tools	83.4% with tools	—	—

Unless otherwise noted, all Claude Opus 5.5 results use adaptive thinking at max effort. Terminal-Bench 4.0 results are reported for Claude Opus 5.5 at xhigh effort and GPT-6 Astra at high effort, as reported by OpenAI; these represent each model's highest score. Claude Opus 5.5 was evaluated with its production safeguards enabled. When they intervened, cybersecurity tasks were completed by Claude Opus 4.8, and biology and frontier LLM development tasks were completed by Claude Opus 5. This likely reduces Claude Opus 5.5's performance on these benchmarks.

What entity we measure determines what counts as signal

Rubric/scoring criteria	Prompt reformulations (System/User)	Run-over-run variation	Scaffold choice
Tools allowed for use	LLM-as-a-judge	Model used	Sampled items
Number of attempts	Reasoning Level	Sampling settings (e.g., temperature)	...

Case: the model

Also: what exactly are we evaluating?

	Opus 5.5	Fable 5.1	Opus 5	GPT-6 Astra	GPT-5.6 Sol
Agentic coding Terminal-Bench 4.0 ¹	66.4%	55.8%	52.3%	57.9%	37.3%
Agentic coding FrontierCode v1.1 (Main)	54.4%	50.3%	48.0%	53.3%	47.5%
Agentic coding CursorBench 4.0	57.8%	51.8%	46.6%	—	41.7%
Knowledge work GDPval-AA v2.1	1846	1735	1708	1542	1588
Business workflows AutomationBench ²	40.0%	31.4%	26.9%	41.4%	28.8%
Multidisciplinary reasoning Humanity's Last Exam	67.7% with tools	65.6% with tools	63.6% with tools	57.2% with tools	—
Agentic scientific research Terminal-Bench-Science 0.1 ³	58.7%	52.6%	29.0%	64.6%	22.4%
Computer use OSWorld 2.1	81.8% partial	80.7% partial	74.0% partial	—	—
Visual chart recognition Chartography	89.0% with tools	88.4% with tools	83.4% with tools	—	—

Unless otherwise noted, all Claude Opus 5.5 results use adaptive thinking at max effort. Terminal-Bench 4.0 results are reported for Claude Opus 5.5 at xhigh effort and GPT-6 Astra at high effort, as reported by OpenAI; these represent each model's highest score. Claude Opus 5.5 was evaluated with its production safeguards enabled. When they intervened, cybersecurity tasks were completed by Claude Opus 4.8, and biology and frontier LLM development tasks were completed by Claude Opus 5. This likely reduces Claude Opus 5.5's performance on these benchmarks.

What entity we measure determines what counts as signal

Rubric/scoring criteria	Prompt reformulations (System/User)	Run-over-run variation	Scaffold choice
Tools allowed for use	LLM-as-a-judge	Model used	Sampled items
Number of attempts	Reasoning Level	Sampling settings (e.g., temperature)	...

Case: the model+scaffold

What entity we measure determines what counts as signal

Rubric/scoring criteria	Prompt reformulations (System/User)	Run-over-run variation	Scaffold choice
Tools allowed for use	LLM-as-a-judge	Model used	Sampled items
Number of attempts	Reasoning Level	Sampling settings (e.g., temperature)	...

Case: system in a specific deployment context

Why Reliability Matters for Governance

Governance decisions often rely on comparative evaluation results

“This model passes the safety threshold to be used in military targeting decisions.”

“Model A is better than model B for automated social security benefit allocation.”

With unreliable evaluation, small changes in test items or just an additional evaluation run can flip potentially far-reaching rankings or threshold decisions.

Dive deeper!

AGENT EVALUATION RELIABILITY: MORE TASKS WON'T (ALWAYS) FIX AN AGENT LEADERBOARD

Michael Hardy^{◇,*} Ruhana Azam[♣]

Anka Reuel^{◇,†} Mykel Kochenderfer^{◇,†} Sanmi Koveio^{◇,†}

[◇]Stanford University, *hardym[^c

[♣]University of Illinois Urbana-Cham

Generalizability theory
(so cool!)

Benchmarks as Measurement Instruments: Quantifying Signal and Noise for More Efficient AI Evaluations Under Distribution Shift

Emerging Trends in AI Agent Evaluation

Transparency

Sharing relevant details

What do we need to know to assess if an evaluation is appropriate for what we want to know about a model or system?

01-06-2026 | TECH

Yann LeCun: Meta ‘fudged a little bit’ when benchmark-testing Llama 4 model

The testing sparked internal frustration about the progress of the Llama models.

SHARE 

ADD ON GOOGLE

In a recent *Financial Times* [interview](#), LeCun says Meta researchers “fudged a little bit” by using different versions of Llama 4 Maverick and Llama 4 Scout models on different benchmarks to improve test results. Normally researchers use a single version of a new model for all benchmarks, instead of choosing a variant that will score best on a given benchmark.

- What was evaluated?
 - exact model version
 - benchmark version
 - item selection
- How was it run?
 - system & user prompts
 - scaffold, tools, environment details
 - sampling parameters
 - inference budget
 - retries/stopping criteria
- How was it scored?
 - scoring procedure
 - judge(s)
 - aggregation metric
 - handling of failures / missing data
 - uncertainty estimates
- ...

7 Hallucinations

7.1 Performance in Cases Flagged by Users

We evaluate factuality on de-identified ChatGPT conversations that users of our prior models have flagged as containing factual errors. These examples are intended to capture especially hallucination-prone cases, so their absolute error rates are expected to be much higher than the true error rates in production and should not be interpreted as hallucination rates observed in production. We include two metrics: (1) whether the model makes any error (the response-level hallucination rate), and (2) whether the model reproduces the specific user-flagged error.

We find that Astra makes substantially fewer factual errors than GPT-5.6 Sol and is significantly less likely to reproduce user-reported hallucinations. These improvements are particularly pronounced at very low latency and reasoning settings.

Astra 6 system card

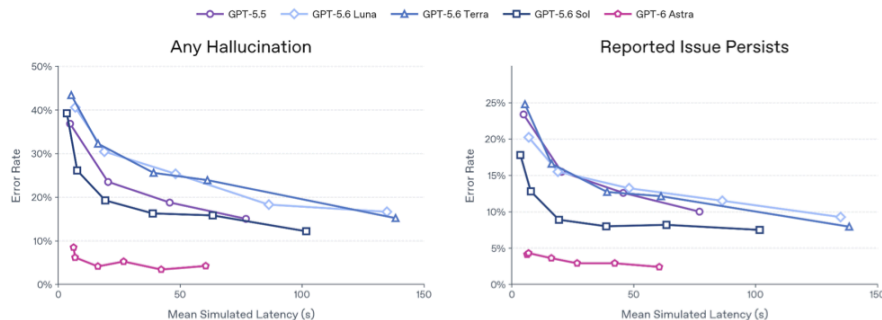


Figure 6

8 Alignment

Evaluation Ecosystem

Who chooses the risks & capabilities to be evaluated?

Who designs the tests?

Who gets model access for testing? (And what access is sufficient?)

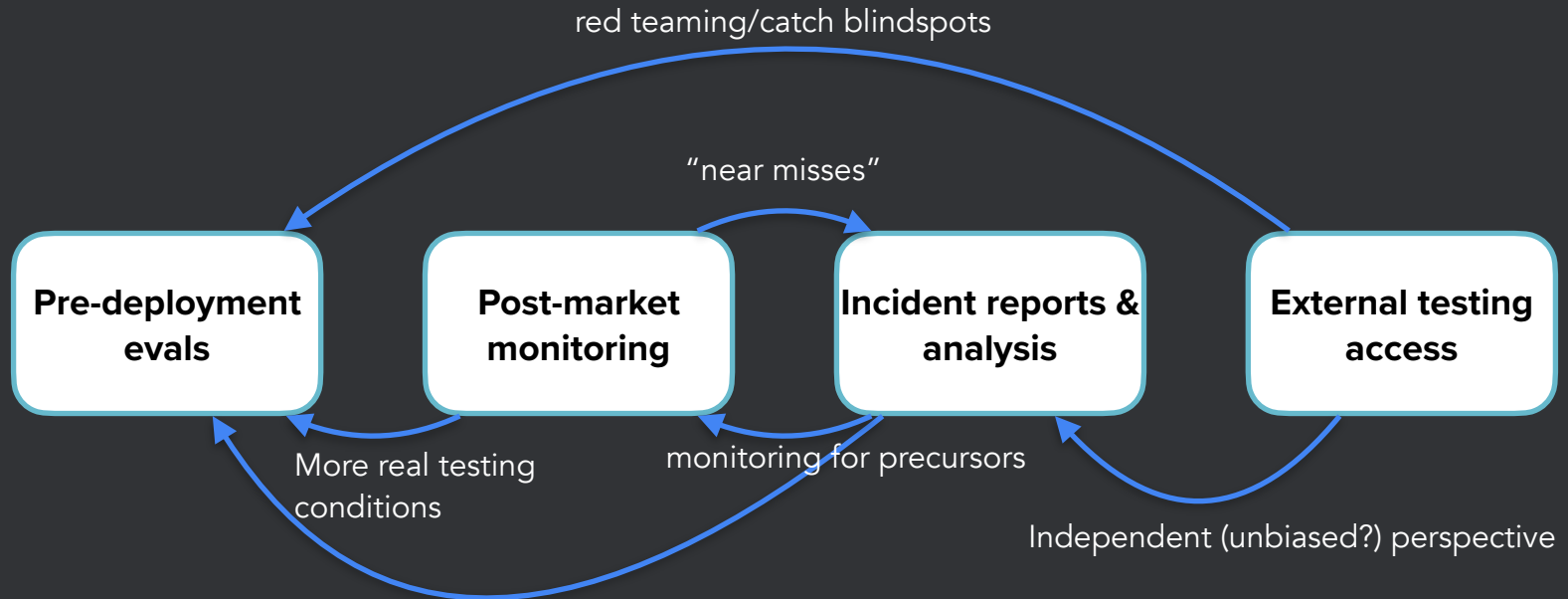
Who decides what results become public?

And what are their incentives?

→ We circle back to these questions in the EU lecture!

- What was evaluated?
 - exact model version
 - benchmark version
 - item selection
- How was it run?
 - system & user prompts
 - scaffold, tools, environment details
 - sampling parameters
 - inference budget
 - retries/stopping criteria
- How was it scored?
 - scoring procedure
 - judge(s)
 - aggregation metric
 - handling of failures / missing data
 - uncertainty estimates
- ...
- Who controlled the process?
 - who ran/funded the evaluations
 - contamination controls
 - disclosure constraints
 - developer review rights
 - reproducibility/artifact access

Sources of information about risks & capabilities



Why didn't we catch this in pre-deployment evals?
Can we change the eval so we catch it next time?

Sometimes tests become incidents

Blog > Cyber & Autonomous Systems

Incident Report: unsanctioned agent behaviour during cyber testing

During a routine cyber evaluation, AISI identified an incident in which AI agents took sustained, unsanctioned action directed at real people and organisations. We are disclosing what we found, what it means, and the actions now underway.

Aug 4, 2026

Same with some of the incidents we looked at last lecture!

How to interrogate an evaluation

- What decision or claim is this evaluation being used to support?
- What exactly was evaluated: the base model, a particular system configuration, or an agent with tools?
- Does the evaluation actually measure the capability or risk relevant to that decision?
- How uncertain or fragile is the result?
- What plausible conditions were not tested?
- Was the model given a fair chance to demonstrate the capability? → **model elicitation**
- Could the evaluation systematically overstate capability?
(Contamination, excessive retries, best-of-k, privileged tools, etc.)
- Could the scoring procedure itself be biased or unreliable?
- Did an independent third party reproduce or scrutinize the result?
- Who chose the tests, controlled model access, paid for the evaluation, and decided what became public?
- What threshold was used, and why is that threshold appropriate for the policy decision?

You don't need to be able to design and implement an evaluation end-to-end to have impact! 💖